AIMD-HF

Educational Ab Initio Molecular Dynamics with Restricted Hartree–Fock / STO-3G

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Abstract

This repository contains a fully self-contained, educational implementation of real-time *Ab Initio Molecular Dynamics* (AIMD) using the Restricted Hartree–Fock (RHF) method with the minimal STO-3G basis set.

The code strongly prioritizes **conceptual clarity and pedagogical value** over computational performance. Every fundamental step of quantum chemistry and molecular simulation is implemented explicitly and clearly commented in a single C++ file.

1 Introduction

AIMD-HF is an interactive educational simulator that couples a full quantum-mechanical electronic structure calculation (Restricted Hartree–Fock) with classical nuclear motion in real time.

Unlike conventional molecular dynamics packages that rely on pre-fitted force fields, this program computes the energy and forces *on-the-fly* by solving the electronic Schrödinger equation approximately via the Hartree–Fock method at every molecular dynamics step.

This allows users to directly visualize the intimate relationship between electronic structure and atomic dynamics.

2 Key Features

- Complete Restricted Hartree–Fock (RHF) implementation with STO-3G basis set
- Explicit evaluation of all one- and two-electron integrals (overlap, kinetic energy, nuclear attraction, and electron repulsion)

- Self-Consistent Field (SCF) procedure with adaptive damping and Jacobi diagonalization
- Nuclear forces computed via central finite differences
- Velocity Verlet integrator (BAOAB splitting)
- Langevin thermostat (friction + stochastic noise)
- Berendsen barostat for pressure control
- Reflective (elastic) boundary conditions in a cubic simulation box
- Real-time 3D visualization using OpenGL and GLUT
- Dynamic chemical bond visualization (yellow cylinders)
- Real-time rolling graph of Hartree–Fock energy
- Comprehensive Heads-Up Display (HUD) showing physical observables
- **Interactive Educational Panels (v1.1)** — extracted at virtually zero extra cost:
  - **[O]** Molecular Orbital Diagram (energies, occupancy, spin arrows, HOMO/LUMO)
  - **[M]** Mulliken Population Analysis (gross population and net atomic charges)
  - **[E]** Energy Decomposition ($T + V_{ne} + V_{ee} + V_{nn}$) with visual bars
  - **[F]** Real-time quantum force vectors displayed as red 3D arrows on each atom
- Multi-threaded design: quantum calculations run in a background thread
- Mutex-protected render buffers for smooth visualization

3 Compilation (Linux)

```
g++ aimd_hf.cpp -o aimd_hf -lGL -lGLU -lglut -lm -lpthread -O2
```

4 Execution

```
./aimd_hf [N]
```

`N` = number of hydrogen atoms (**must be even** for closed-shell RHF). Default value is 2.

5 Keyboard Controls

Key	Action
W / S	Zoom camera in / out
A / D	Move camera left / right
Z / X	Move camera up / down
+ / -	Increase / decrease target temperature (±50 K)
O	Toggle Molecular Orbital Diagram
M	Toggle Mulliken Population Analysis
E	Toggle Energy Decomposition Panel
F	Toggle Force Arrows (red 3D vectors)
ESC	Exit the program

6 Educational Value

This project was designed as a powerful teaching tool for quantum chemistry, computational chemistry, and molecular simulation courses. Users can visually explore and understand:

- SCF convergence behavior during real-time molecular dynamics
- The direct coupling between electronic structure and nuclear motion
- How temperature affects both molecular geometry and electronic properties
- Fluctuations in Mulliken atomic charges as atoms move
- Evolution of the HOMO–LUMO gap throughout the simulation
- Real quantum mechanical forces acting on each atom (visualized in real time)
- The relative contribution of each energy component (kinetic, nuclear attraction, electron-electron repulsion, and nuclear repulsion)
- The practical effects of numerical approximations in quantum chemistry

7 Design Philosophy

- **Clarity over performance**: All core algorithms are written explicitly
- Every major step (integral evaluation, SCF, force calculation, MD integration) is clearly commented
- Minimal use of external libraries (only OpenGL/GLUT for visualization)
- Strong emphasis on pedagogical value rather than scalability
- Ideal for students, instructors, and researchers who want to understand what happens "under the hood" in ab initio simulations

8 Limitations

- $\mathcal{O}(N^4)$ scaling due to explicit computation of two-electron integrals
- Forces are calculated numerically (6N + 1 SCF evaluations per MD step)
- Practical limit of approximately 12 hydrogen atoms
- STO-3G is a minimal basis set — results are qualitative and intended for education only
- Not suitable for production-level scientific simulations or large systems

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Repository: https://github.com/gallactuz/AIMD-HF